

Q1 Implement a MPI program to read N values in the root process. Root process sends one value to each process. Every process receives it and finds the factorial of that number and returns it to the root process. Root process gathers the factorial and finds sum of it. Use N number of processes.

```
SCSED1@dsl-11:~/Documents/240905318_Anivin_PCAP/week5$ mpirun -np 4 ./pgm1
Enter 4 values:
1
2
3
4
Factorials:
1 2 6 24
Sum of factorials = 33
```

Q2 Implement a MPI program to read an integer value M and NXM elements into an ID array in the root process, where N is the number of processes. Root process sends M elements to each process. Each process finds average of M elements it received and sends these average values to root. Root collects all the values and finds the total average. Use collective communication routines.

```
SCSED1@dsl-11:~/Documents/240905318_Anivin_PCAP/week5$ mpicc pgm2.c -o pgm2
SCSED1@dsl-11:~/Documents/240905318_Anivin_PCAP/week5$ mpirun -np 5 ./pgm2
Enter M: 4
Enter 20 elements:
1
2
3
4
5
6
7
8
9
0
11
1
2
15
199
10
112
187
156
100
Total average = 41.90
```

Q3 Implement a MPI program to read a string. Using N processes (string length is evenly divisible by N), find the number of non-vowels in the string. In the root process print number of non-vowels found by each process and print the total number of non-vowels.

```
SCSED1@dsl-11:~/Documents/240905318_Anivin_PCAP/week5$ mpicc pgm3.c -o pgm3
SCSED1@dsl-11:~/Documents/240905318_Anivin_PCAP/week5$ mpirun -np 5 ./pgm3
Enter string: Anirvin
Process 0: 1 non-vowels
Process 1: 1 non-vowels
Process 2: 0 non-vowels
Process 3: 1 non-vowels
Process 4: 1 non-vowels
Total non-vowels = 4
```

Q4 Implement a MPI program to read two strings S1 and S2 of same length in the root process. Using N processes including the root (string length is evenly divisible by N), produce the resultant string as shown below. Use collective communication routines. Display the resultant string in the root process.

Example:

String S1: string

String S2: length

Resultant String: slternitgth

```
SCSED1@dsl-11: ~/Documents/240905318_Anivin_PCAP/week5$ mpirun -np 5 ./pgm4
Enter S1: hello
Enter S2: 12345
Resultant String: h1e2l3l4o5
```

Q5 Implement parallel selection sort [hint: Use merge sort after sorting subarrays using parallel selection sort]

```
SCSED1@dsl-11: ~/Documents/240905318_Anivin_PCAP/week5$ mpirun -np 5 ./pgm5
Enter number of elements: 10
Enter elements:
6
11
901
43
101
455
16
299
150
151
Sorted array:
6 11 16 43 101 150 151 299 455 901
```